

Predictors of Absolute VO2max

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Setup

Combine Raw Data Files

This step was run once to combine the individual Excel files into `combined_tests.csv` and is not re-run after.

Data Cleaning

```
df_raw <- read.csv(here("data", "combined_tests.csv"))

df_raw <- df_raw |>
  rename(file = "source_file")

df_raw_sum <- read.csv(here("data", "Data_Summary_with_age.csv"))

df_raw_sum <- df_raw_sum |>
  select("file", "age")

### Making a df that only has variables that are needed for the study

colnames(df_raw)

df_clean <- df_raw |>
  select("file", "gender", "event", "test", "height", "weight", "fat", "lbm", "hb",
        "hct", "vo2max", "vo2max_absolute", "hrmax", "load_tlac",
        "hr_tlac", "percent_vo2max_tlac", "percent_h_rmax_tlac", "bla_tlac")

df_clean <- merge(df_clean, df_raw_sum, by = "file")

df_clean <- df_clean |>
  relocate("age", .after = gender)

df_clean <- df_clean |>
  distinct(file, .keep_all = T)

df_clean <- df_clean |>
  filter(!file %in% c("A539.xlsx", "A140.xlsx", "A562.xlsx", "A1065.xlsx"))

### Df manipulation and cleaning for analysis
colnames(df_clean)

anyNA(df_clean)

colSums(is.na(df_clean))

df_clean$weight[df_clean$weight == "Missing"] <- NA

df_clean <- df_clean |>
  na.omit() |>
  mutate(weight = as.numeric(as.character(weight)))

df_clean <- df_clean |>
  mutate(across(where(is.numeric), ~ round(., 2)))

df_clean <- df_clean |>
  filter(age >= 18)

df_clean <- df_clean |>
  rename(sex = gender)
```

```
df_clean <- df_clean %>%
  mutate(test = case_match(test,
    c("Rowing", "Kayak") ~ "watersports",
    .default = test
  ))
```

Cleaning notes

- Four participants (A539, A140, A562, A1065) were excluded because of implausible or irreconcilable values (VO2max, body fat %, hematocrit).
- Participants under 18 or with missing values were removed, and rowing and kayak were combined into “watersports”.
- Final analytic sample: $N = 711$ (151 women, 560 men).

Descriptive Statistics

```
# Overall
df_clean |>
  summarise(
    n = n(),
    age_mean = mean(age),
    age_sd = sd(age),
    abs_vo2_mean = mean(vo2max_absolute),
    abs_vo2_sd = sd(vo2max_absolute)
  ) |>
  knitr::kable(digits = 2, caption = "Overall sample")
```

Table 1: Overall sample

n	age_mean	age_sd	abs_vo2_mean	abs_vo2_sd
711	27.55	7.01	4.61	0.86

```
# By modality
df_clean |>
  group_by(test) |>
  summarise(
    n = n(),
    age_mean = mean(age),
    age_sd = sd(age),
    abs_vo2_mean = mean(vo2max_absolute),
    abs_vo2_sd = sd(vo2max_absolute),
    .groups = "drop"
  ) |>
  knitr::kable(digits = 2, caption = "By modality")
```

Table 2: By modality

test	n	age_mean	age_sd	abs_vo2_mean	abs_vo2_sd
Cycling	253	30.29	6.92	4.58	0.74
Running	230	29.37	6.97	4.26	0.88
watersports	228	22.67	3.97	4.99	0.81

```
# By sex
df_clean |>
  group_by(sex) |>
  summarise(
    n = n(),
    age_mean = mean(age),
    age_sd = sd(age),
    abs_vo2_mean = mean(vo2max_absolute),
    abs_vo2_sd = sd(vo2max_absolute),
    .groups = "drop"
  ) |>
  knitr::kable(digits = 2, caption = "By sex")
```

Table 3: By sex

sex	n	age_mean	age_sd	abs_vo2_mean	abs_vo2_sd
F	151	28.18	6.09	3.40	0.49
M	560	27.38	7.23	4.93	0.62

```
# Check that there are enough participants in each moderator group
df_clean |>
  count(sex) |>
  mutate(percent = round(n / sum(n) * 100, 2)) |>
  knitr::kable(caption = "Sex distribution")
```

Table 4: Sex distribution

sex	n	percent
F	151	21.24
M	560	78.76

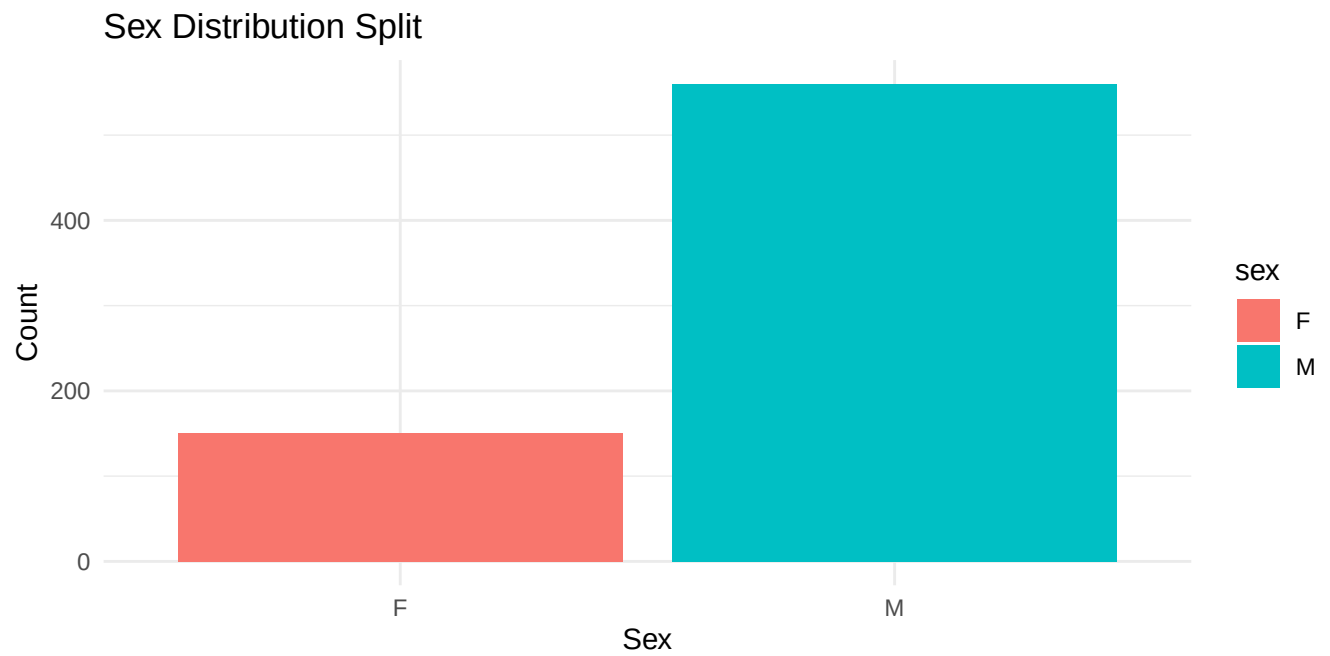
```
df_clean |>
  count(test) |>
  mutate(percent = round(n / sum(n) * 100, 2)) |>
  knitr::kable(caption = "Modality distribution")
```

Table 5: Modality distribution

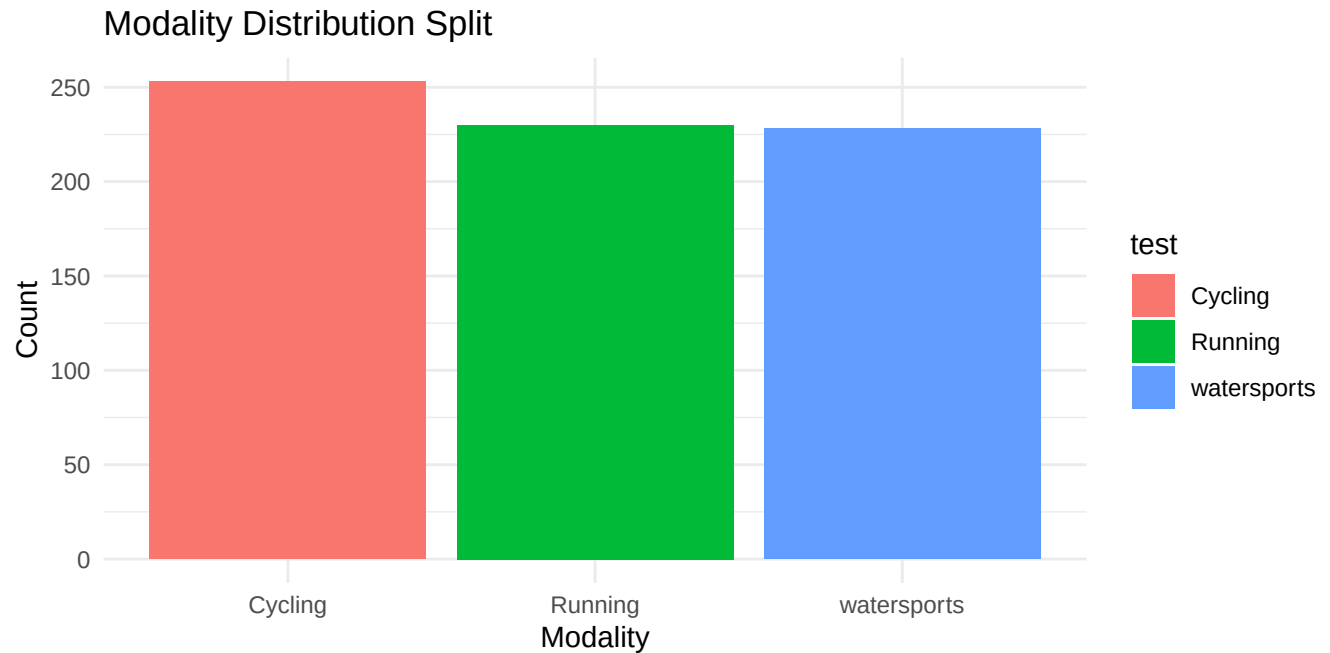
test	n	percent
Cycling	253	35.58

test	n	percent
Running	230	32.35
watersports	228	32.07

```
ggplot(df_clean, aes(x = sex, fill = sex)) +
  geom_bar() +
  labs(title = "Sex Distribution Split", y = "Count", x = "Sex") +
  theme_minimal()
```



```
ggplot(df_clean, aes(x = test, fill = test)) +
  geom_bar() +
  labs(title = "Modality Distribution Split", y = "Count", x = "Modality") +
  theme_minimal()
```



Research Question 1

How well do the known predictors (body fat %, lean body mass, hemoglobin, hematocrit) jointly explain absolute VO2max, without moderators? Body weight is left out because it is almost fully determined by fat % and lean mass.

Model and coefficients

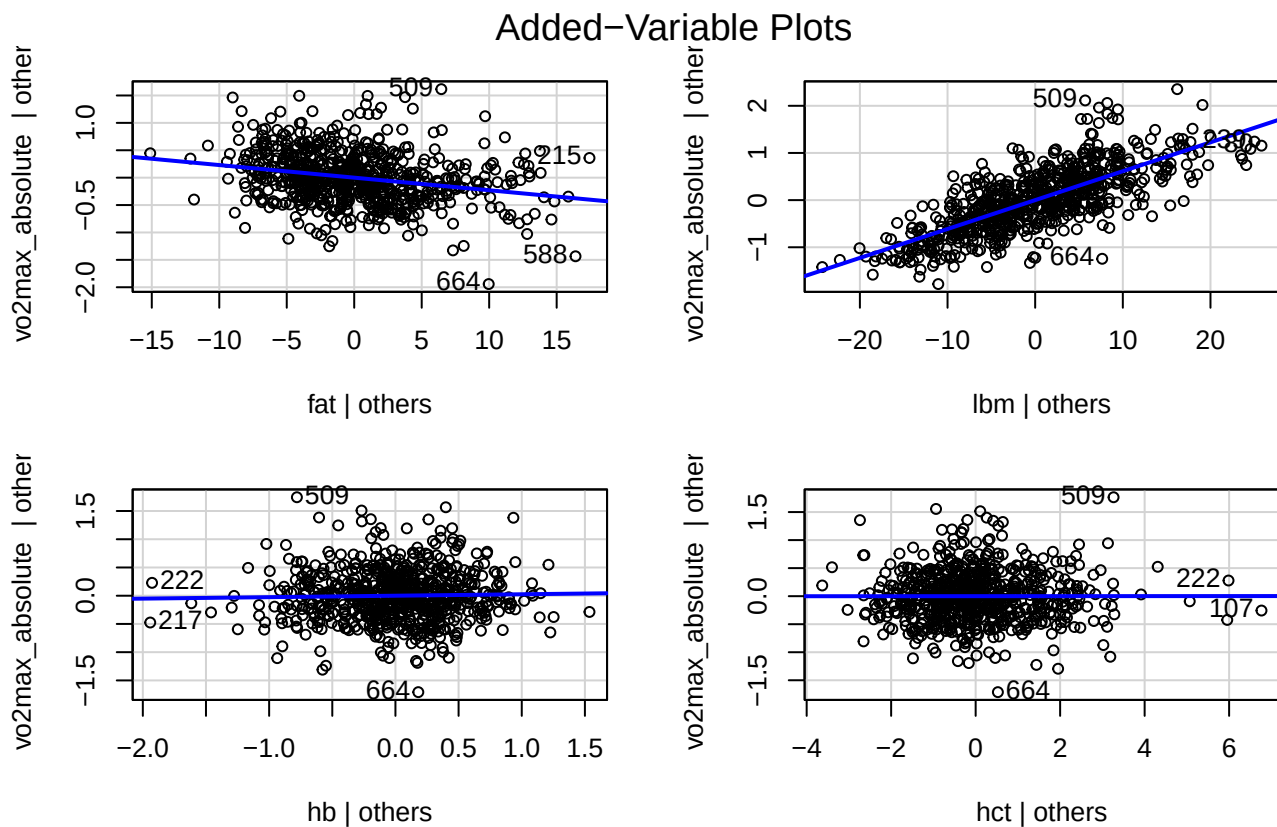
```
A1 <- lm(vo2max_absolute ~ fat + lbm + hb + hct, data = df_clean)
```

```
summary(A1)
```

```
##
## Call:
## lm(formula = vo2max_absolute ~ fat + lbm + hb + hct, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.71254 -0.29066 -0.01944  0.26123  1.76367
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.7139775  0.3051941   2.339   0.0196 *
## fat         -0.0229935  0.0033994  -6.764 2.82e-11 ***
## lbm          0.0612839  0.0020766  29.512 < 2e-16 ***
## hb           0.0256867  0.0363270   0.707   0.4797
## hct          0.0002442  0.0128642   0.019   0.9849
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4546 on 706 degrees of freedom
## Multiple R-squared:  0.7224, Adjusted R-squared:  0.7208
## F-statistic: 459.3 on 4 and 706 DF,  p-value: < 2.2e-16
```

```
avPlots(A1)
```



```
tidy(A1, conf.int = TRUE) |> tab("RQ1: Regression coefficients")
```

Table 6: RQ1: Regression coefficients

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	0.714	0.305	2.339	0.020	0.115	1.313
fat	-0.023	0.003	-6.764	<0.001	-0.030	-0.016
lbm	0.061	0.002	29.512	<0.001	0.057	0.065
hb	0.026	0.036	0.707	0.480	-0.046	0.097
hct	0.000	0.013	0.019	0.985	-0.025	0.026

```
glance(A1) |>
  select(r.squared, adj.r.squared, sigma, statistic, p.value, nobs) |>
  tab("RQ1: Overall model fit")
```

Table 7: RQ1: Overall model fit

r.squared	adj.r.squared	sigma	statistic	p.value	nobs
0.722	0.721	0.455	459.32	<0.001	711

Interpretation: Together the four predictors explain about 72% of the variance in absolute VO2max ($R^2 = 0.72$). Lean mass is the strongest predictor: each additional kg is associated with about 0.061 L/min more VO2max (95% CI 0.057 to 0.065). Each additional percentage point of body fat is associated with about 0.023 L/min less (CI -0.030 to -0.016). Hemoglobin and hematocrit are not associated with VO2max once body composition is accounted for (their CIs include zero).

Multicollinearity

```
vif(A1) |>
  enframe(name = "predictor", value = "VIF") |>
  knitr::kable(digits = 2, caption = "RQ1: Variance inflation factors")
```

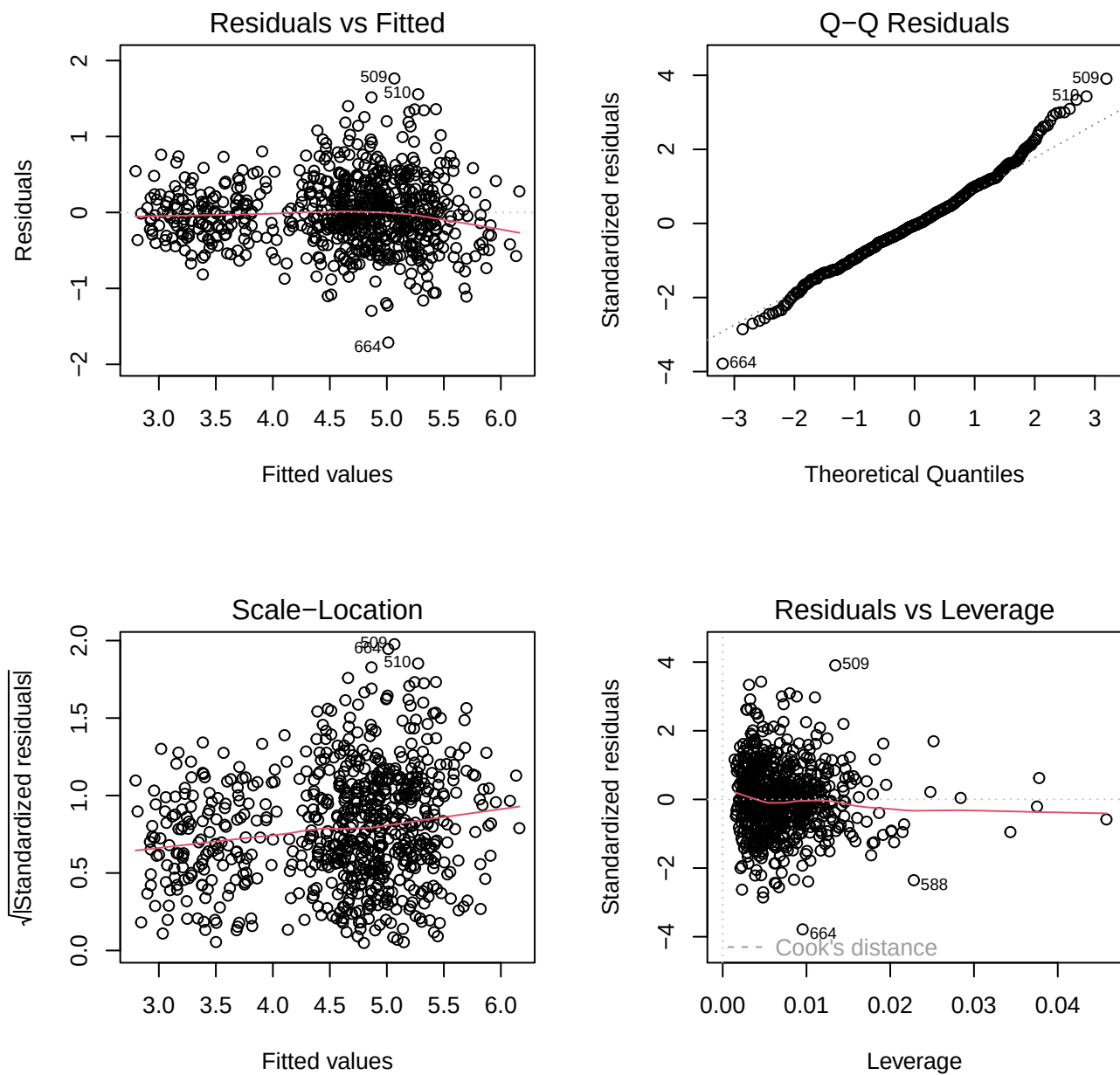
Table 8: RQ1: Variance inflation factors

predictor	VIF
fat	1.40
lbm	1.60
hb	4.90
hct	4.71

Interpretation: Fat % (1.4) and lean mass (1.6) show little collinearity. Hemoglobin (4.9) and hematocrit (4.7) are correlated with each other, as expected, so their individual coefficients are less precise. This does not affect the joint test of the blood variables below.

Assumption checks

```
par(mfrow = c(2, 2)); plot(A1); par(mfrow = c(1, 1))
```



```
bptest(A1) |> tidy() |> tab("Breusch-Pagan test for heteroscedasticity")
```

Table 9: Breusch-Pagan test for heteroscedasticity

statistic	p.value	parameter	method
25.065	<0.001	4	studentized Breusch-Pagan test

Interpretation: The residuals are centered around zero with no curvature, and the Q-Q plot follows the line. No observation has a large Cook's distance. The Breusch-Pagan test detects some non-constant variance

($p < 0.001$), so robust standard errors are used next as a check.

Robust standard errors

```
coeftest(A1, vcov = vcovHC(A1, type = "HC3")) |>
  tidy() |>
  tab("RQ1: Coefficients with robust (HC3) standard errors")
```

Table 10: RQ1: Coefficients with robust (HC3) standard errors

term	estimate	std.error	statistic	p.value
(Intercept)	0.714	0.334	2.135	0.033
fat	-0.023	0.004	-6.167	<0.001
lbm	0.061	0.002	29.594	<0.001
hb	0.026	0.036	0.721	0.471
hct	0.000	0.013	0.019	0.985

Interpretation: The robust standard errors are almost identical to the regular ones (e.g. lean mass 0.0021 in both), and the conclusions do not change: lean mass and fat % are significant, hemoglobin and hematocrit are not. The heteroscedasticity is therefore not a concern for these results.

Do the blood variables add anything?

```
A1_body <- lm(vo2max_absolute ~ fat + lbm, data = df_clean)

anova(A1_body, A1) |>
  tidy() |>
  mutate(model = c("fat + lbm", "fat + lbm + hb + hct"), .before = 1) |>
  select(-term) |>
  tab("Model comparison: body composition only vs. adding hb and hct")
```

Table 11: Model comparison: body composition only vs. adding hb and hct

model	df.residual	rss	df	sumsq	statistic	p.value
fat + lbm	708	146.254	NA	NA	NA	NA
fat + lbm + hb + hct	706	145.872	2	0.382	0.924	0.397

Interpretation: Adding hemoglobin and hematocrit to a model with fat % and lean mass does not improve fit ($F = 0.92$, $p = 0.40$). Body composition accounts for the explainable variance, and the blood variables add essentially nothing.

Research Question 2

Do age, sex, or exercise modality moderate the associations between the known predictors and absolute VO2max?

Models

```
df_clean <- df_clean |>
  mutate(sex = factor(sex), test = factor(test))

# For each moderator: a main-effects model and a full model with all predictor x moderator interactions
A2_age_main <- lm(vo2max_absolute ~ fat + lbm + hb + hct + age, data = df_clean)
A2_age      <- lm(vo2max_absolute ~ (fat + lbm + hb + hct) * age, data = df_clean)

A2_sex_main <- lm(vo2max_absolute ~ fat + lbm + hb + hct + sex, data = df_clean)
A2_sex      <- lm(vo2max_absolute ~ (fat + lbm + hb + hct) * sex, data = df_clean)

A2_test_main <- lm(vo2max_absolute ~ fat + lbm + hb + hct + test, data = df_clean)
A2_test      <- lm(vo2max_absolute ~ (fat + lbm + hb + hct) * test, data = df_clean)

# Model fit comparison
bind_rows(
  "RQ1 (no moderator)" = glance(A1),
  "Age"                = glance(A2_age),
  "Sex"                = glance(A2_sex),
  "Modality"           = glance(A2_test),
  .id = "model"
) |>
  select(model, r.squared, adj.r.squared, sigma) |>
  tab("R-squared with and without moderators")
```

Table 12: R-squared with and without moderators

model	r.squared	adj.r.squared	sigma
RQ1 (no moderator)	0.722	0.721	0.455
Age	0.735	0.732	0.446
Sex	0.729	0.726	0.450
Modality	0.735	0.730	0.447

```
tidy(A2_age) |> tab("Full model: predictors x age")
```

Table 13: Full model: predictors x age

term	estimate	std.error	statistic	p.value
(Intercept)	-2.193	1.265	-1.734	0.083
fat	0.017	0.015	1.129	0.259
lbm	0.069	0.008	8.146	<0.001
hb	0.162	0.136	1.184	0.237
hct	0.004	0.049	0.084	0.933
age	0.105	0.044	2.380	0.018
fat:age	-0.001	0.001	-2.716	0.007
lbm:age	0.000	0.000	-0.958	0.338
hb:age	-0.005	0.005	-1.017	0.309
hct:age	0.000	0.002	-0.081	0.935

```
tidy(A2_sex) |> tab("Full model: predictors x sex")
```

Table 14: Full model: predictors x sex

term	estimate	std.error	statistic	p.value
(Intercept)	0.155	0.644	0.241	0.810
fat	-0.002	0.007	-0.287	0.775
lbm	0.064	0.007	9.616	<0.001
hb	0.129	0.078	1.661	0.097
hct	-0.036	0.027	-1.366	0.172
sexM	1.324	0.765	1.730	0.084
fat:sexM	-0.024	0.008	-2.840	0.005
lbm:sexM	-0.008	0.007	-1.063	0.288
hb:sexM	-0.142	0.088	-1.610	0.108
hct:sexM	0.041	0.030	1.365	0.173

```
tidy(A2_test) |> tab("Full model: predictors x modality")
```

Table 15: Full model: predictors x modality

term	estimate	std.error	statistic	p.value
(Intercept)	2.541	0.519	4.896	<0.001
fat	-0.042	0.006	-7.068	<0.001
lbm	0.054	0.004	14.920	<0.001
hb	-0.063	0.064	-0.971	0.332
hct	0.005	0.023	0.217	0.828
testRunning	-1.947	0.750	-2.595	0.010
testwatersports	-3.567	0.819	-4.352	<0.001
fat:testRunning	0.015	0.009	1.727	0.085
fat:testwatersports	0.039	0.009	4.593	<0.001
lbm:testRunning	0.007	0.005	1.225	0.221
lbm:testwatersports	0.013	0.006	2.303	0.022
hb:testRunning	0.080	0.089	0.901	0.368
hb:testwatersports	0.195	0.089	2.197	0.028
hct:testRunning	0.001	0.032	0.046	0.964
hct:testwatersports	-0.018	0.031	-0.587	0.557

Interpretation: Adding the moderator terms raises R^2 only slightly, from 0.722 in RQ1 to 0.735 (age), 0.730 (sex) and 0.735 (modality). The predictors are not centered, so the main-effect rows (and the sex and modality rows) describe the reference case where the other predictors equal zero and should not be interpreted on their own. The rows to read are the interaction terms (e.g. **fat:age**), which show how a predictor's slope differs by age, sex or modality. The tests below make that clearer.

Step 1: Does each moderator change the relationships at all?

Joint F-test of all four predictor x moderator interactions, with Holm adjustment across the three moderators.

```
bind_rows(
  Age      = tidy(anova(A2_age_main, A2_age)),
  Sex      = tidy(anova(A2_sex_main, A2_sex)),
  Modality = tidy(anova(A2_test_main, A2_test)),
  .id = "moderator"
) |>
  filter(!is.na(statistic)) |>
  select(moderator, df, F_value = statistic, p.value) |>
  mutate(p_holm = p.adjust(p.value, method = "holm")) |>
  tab("Joint test of all interactions per moderator")
```

Table 16: Joint test of all interactions per moderator

moderator	df	F_value	p.value	p_holm
Age	4	2.465	0.044	0.044
Sex	4	2.943	0.020	0.040
Modality	8	3.794	<0.001	<0.001

Interpretation: All three moderators change the predictor relationships, but with very different strength. Modality is clearly significant ($F = 3.79$, Holm-adjusted $p = 0.0007$). Age ($p = 0.044$) and sex ($p = 0.040$) are only just significant after adjustment, so those results are suggestive rather than strong.

Step 2: Which predictors are moderated?

```
Anova(A2_age, type = 2) |> tidy() |> tab("Type II tests: predictors x age")
```

Table 17: Type II tests: predictors x age

term	sumsq	df	statistic	p.value
fat	8.756	1	44.080	<0.001
lbm	174.095	1	876.438	<0.001
hb	0.119	1	0.598	0.440
hct	0.000	1	0.000	0.984
age	4.667	1	23.497	<0.001
fat:age	1.465	1	7.374	0.007
lbm:age	0.182	1	0.918	0.338
hb:age	0.206	1	1.035	0.309
hct:age	0.001	1	0.007	0.935
Residuals	139.246	701	NA	NA

```
Anova(A2_sex, type = 2) |> tidy() |> tab("Type II tests: predictors x sex")
```

Table 18: Type II tests: predictors x sex

term	sumsq	df	statistic	p.value
fat	4.126	1	20.342	<0.001
lbm	98.602	1	486.181	<0.001
hb	0.053	1	0.261	0.610
hct	0.025	1	0.123	0.726
sex	1.315	1	6.484	0.011
fat:sex	1.636	1	8.066	0.005
lbm:sex	0.229	1	1.130	0.288
hb:sex	0.526	1	2.593	0.108
hct:sex	0.378	1	1.863	0.173
Residuals	142.169	701	NA	NA

```
Anova(A2_test, type = 2) |> tidy() |> tab("Type II tests: predictors x modality")
```

Table 19: Type II tests: predictors x modality

term	sumsq	df	statistic	p.value
fat	9.023	1	45.087	<0.001
lbm	132.694	1	663.026	<0.001
hb	0.166	1	0.829	0.363
hct	0.002	1	0.008	0.930
test	0.504	2	1.259	0.285
fat:test	4.259	2	10.639	<0.001
lbm:test	1.079	2	2.695	0.068
hb:test	0.984	2	2.459	0.086
hct:test	0.103	2	0.258	0.772
Residuals	139.293	696	NA	NA

```
# Direction and size of the age effect from the main-effects model
tidy(A2_age_main, conf.int = TRUE) |>
  filter(term == "age") |>
  tab("Age effect on VO2max (L/min per year), main-effects model")
```

Table 20: Age effect on VO2max (L/min per year), main-effects model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
age	-0.012	0.002	-4.827	<0.001	-0.016	-0.007

Interpretation: The moderation is concentrated in one predictor: body fat. The fat slope differs by age ($p = 0.007$), by sex ($p = 0.005$) and strongly by modality ($p < 0.0001$). The lean mass and blood-variable interactions are not significant (lean mass x modality $p = 0.068$ and hemoglobin x modality $p = 0.086$ are trends only). Age also has an independent main effect on VO2max ($p < 0.001$); the last table gives its direction and size per year.

Robust standard errors

```
coeftest(A2_age, vcov = vcovHC(A2_age, type = "HC3")) |> tidy() |> tab("Robust (HC3): predictors x age")
```

Table 21: Robust (HC3): predictors x age

term	estimate	std.error	statistic	p.value
(Intercept)	-2.193	1.348	-1.627	0.104
fat	0.017	0.014	1.193	0.233
lbm	0.069	0.008	8.520	<0.001
hb	0.162	0.141	1.148	0.251
hct	0.004	0.048	0.084	0.933
age	0.105	0.049	2.145	0.032
fat:age	-0.001	0.001	-2.792	0.005
lbm:age	0.000	0.000	-0.994	0.320
hb:age	-0.005	0.005	-1.007	0.314
hct:age	0.000	0.002	-0.086	0.931

```
coeftest(A2_sex, vcov = vcovHC(A2_sex, type = "HC3")) |> tidy() |> tab("Robust (HC3): predictors x sex")
```

Table 22: Robust (HC3): predictors x sex

term	estimate	std.error	statistic	p.value
(Intercept)	0.155	0.500	0.310	0.757
fat	-0.002	0.006	-0.333	0.739
lbm	0.064	0.006	11.204	<0.001
hb	0.129	0.051	2.502	0.013
hct	-0.036	0.018	-2.067	0.039
sexM	1.324	0.691	1.915	0.056
fat:sexM	-0.024	0.008	-3.008	0.003
lbm:sexM	-0.008	0.006	-1.202	0.230
hb:sexM	-0.142	0.068	-2.089	0.037
hct:sexM	0.041	0.024	1.743	0.082

```
coeftest(A2_test, vcov = vcovHC(A2_test, type = "HC3")) |> tidy() |> tab("Robust (HC3): predictors x modality")
```

Table 23: Robust (HC3): predictors x modality

term	estimate	std.error	statistic	p.value
(Intercept)	2.541	0.555	4.580	<0.001
fat	-0.042	0.006	-6.812	<0.001
lbm	0.054	0.004	14.764	<0.001
hb	-0.063	0.060	-1.035	0.301
hct	0.005	0.020	0.245	0.807
testRunning	-1.947	0.710	-2.741	0.006
testwatersports	-3.567	0.992	-3.595	<0.001
fat:testRunning	0.015	0.008	1.881	0.060

term	estimate	std.error	statistic	p.value
fat:testwatersports	0.039	0.010	3.942	<0.001
lbm:testRunning	0.007	0.005	1.371	0.171
lbm:testwatersports	0.013	0.007	2.050	0.041
hb:testRunning	0.080	0.076	1.053	0.293
hb:testwatersports	0.195	0.094	2.070	0.039
hct:testRunning	0.001	0.026	0.056	0.955
hct:testwatersports	-0.018	0.034	-0.544	0.587

Interpretation: The robust results agree with the F-tests: **fat:age** ($p = 0.005$), **fat:sexM** ($p = 0.003$) and **fat:testwatersports** ($p < 0.001$) remain significant. A few weaker interactions (hemoglobin x sex, lean mass and hemoglobin x watersports) reach $p < 0.05$ only with robust errors, so they are less reliable and should be treated as exploratory.

Step 3: Body fat slope within each group

```
emtrends(A2_test, ~ test, var = "fat") |> as.data.frame() |> tab("Fat slope by modality")
```

Table 24: Fat slope by modality

test	fat.trend	SE	df	lower.CL	upper.CL
Cycling	-0.042	0.006	696	-0.053	-0.030
Running	-0.026	0.007	696	-0.039	-0.014
watersports	-0.002	0.006	696	-0.014	0.010

```
emtrends(A2_sex, ~ sex, var = "fat") |> as.data.frame() |> tab("Fat slope by sex")
```

Table 25: Fat slope by sex

sex	fat.trend	SE	df	lower.CL	upper.CL
F	-0.002	0.007	701	-0.015	0.011
M	-0.026	0.005	701	-0.035	-0.016

```
emtrends(A2_age, ~ age, var = "fat", at = list(age = c(20, 30, 40))) |>
  as.data.frame() |> tab("Fat slope by age")
```

Table 26: Fat slope by age

age	fat.trend	SE	df	lower.CL	upper.CL
20	-0.012	0.005	701	-0.022	-0.002
30	-0.026	0.004	701	-0.033	-0.019
40	-0.040	0.007	701	-0.054	-0.025

Interpretation: These are the effects of one additional percentage point of body fat on VO2max (L/min) within each group. Body fat lowers VO2max most in cyclists (about -0.04) and less in runners (about -0.03),

while in watersports athletes the slope is close to zero. Among men the fat slope is about -0.026, while among women it is close to zero. The fat penalty also grows with age (more negative at 40 than at 20). Check the confidence intervals: slopes whose CIs include zero mean body fat is not clearly related to VO2max in that group.

Summary

- Lean mass (positive) and body fat % (negative) explain about 72% of the variance in absolute VO2max. Hemoglobin and hematocrit add nothing beyond them.
- Age, sex and modality moderate these relationships mainly through body fat. The modality effect is the strongest, and the age and sex effects are more modest.
- Age, sex and modality differ from each other in this sample (watersports athletes are younger and mostly male), so the three fat interactions may partly reflect the same underlying difference.